The Boundaries of Inclusive Growth: Factor-Biased Technical Change Suitability, Human Capital, and Institutional Quality

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Abstract

Does the alignment between a country's direction of technical change and its factor endowments reduce income inequality, and under what conditions? Using a balanced panel of 64 economies from 1980 to 2019, we show that greater capital-biased technical change suitability significantly reduces the log Gini coefficient by approximately 0.82 percent per standard deviation. This equalizing effect, however, is conditional on two boundary factors. Human capital substantially amplifies the suitability–inequality channel: a one-standard-deviation increase in human capital raises the marginal equalizing effect by roughly 2.1 percentage points, operating primarily through global value chain upgrading and high-quality job creation. Institutional quality reduces inequality

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directly but acts as a threshold condition rather than a linear amplifier: only when both institutions and human capital are simultaneously weak is the equalizing effect of suitability significantly attenuated. These findings imply that technology-factor alignment must be accompanied by investment in people and governance capacity to generate broad-based growth.

JEL: O33, D31, J24, P16, F66

Keywords: biased technical change, income inequality, human capital, institutional quality, global value chains, factor endowments

1 Introduction

A central puzzle in development economics concerns why technological progress produces markedly different distributional outcomes across countries. Capital-intensive technologies imported by capital-scarce economies generate weaker employment and wage gains than identical technologies operating in capital-abundant settings (Acemoglu and Zilibotti, 2001; Caselli and Coleman, 2006). Yet even after accounting for factor endowment mismatches, the equalizing effects of technically appropriate innovation remain heterogeneous. Some economies successfully translate factor-aligned technical change into broad-based income gains; others see productivity improvements appropriated by narrow elites. Understanding what determines which outcome prevails is central to designing technology policy that promotes inclusive growth.

This paper argues that two boundary conditions mediate whether capital-biased technical change suitability reduces income inequality: *human capital* and *institutional quality*. When the labor force is insufficiently educated, capital-complementary technologies raise productivity without expanding the set of tasks workers can perform alongside machines. The distributional benefits of technical change then accrue primarily to capital owners and a thin stratum of high-skill workers, leaving the broad middle of the wage distribution unaffected or worse off (Krusell et al., 2000; Moll et al., 2022). When institutions are weak, efficiency gains are captured by politically connected insiders through rent-seeking rather than diffused via competitive labor markets and public goods provision (Acemoglu et al., 2005; Hall and Jones, 1999). Both conditions choke off the channels through which factor-aligned technical change would otherwise compress wage dispersion.

We test this mechanism using variation in *capital-biased technical change suitability*—the degree to which an economy's direction of technical change aligns with its capital-labor ratio—across 64 economies from 1980 to 2019. Our suitability measure follows Li et al. (2026) and Li et al. (2024), who construct suitability as the log ratio of the marginal products of capital and labor, a sufficient statistic for factor-direction alignment under standard constant-returns production assumptions. Higher suitability means that the prevailing direction of capital-biased technical progress is more consistent with the economy's relative factor abundance, reducing the implicit wedge between technology and endowments identified by Acemoglu and Zilibotti (2001).

Our empirical strategy exploits two-way fixed effects (TWFE) with Driscoll–Kraay standard errors (Driscoll and Kraay, 1998) that are robust to cross-sectional dependence and serial correlation, which are pervasive in cross-country panel data. We augment the TWFE baseline with interaction terms between suitability and (i) institutional quality, measured by the Fraser Institute's Economic Freedom Index (Gwartney et al., 2020), and (ii) human capital, proxied by the log of mean schooling in 1970 (Barro and Lee, 1996). The pre-sample human capital proxy reduces concerns about reverse causation from contemporary inequality to schooling investment.

The paper yields four main results. First, technical change suitability significantly reduces income inequality in the baseline specification. A one-standard-deviation increase in suitability (0.48 log points) is associated with a 0.82 percent decline in the log Gini coefficient, with the coefficient -0.017 significant at the 5 percent level. This confirms the aggregate finding in Li et al. (2026) and establishes it in the moderation framework of this paper.

Second, human capital is quantitatively the most important amplifier of the suitability–inequality relationship. The interaction between suitability and 1970 human capital carries a coefficient of -0.038 (significant at 1 percent), implying that moving from the median to one standard deviation above median human capital raises the equalizing effect of a one-unit increase in suitability by 2.1 percentage points. A high-human-capital dummy interaction confirms this nonlinearity.

Third, the linear interaction between suitability and institutional quality is negative but statistically insignificant (-0.004, $p > 0.10$). Institutions reduce inequality on their own, but they do not monotonically amplify the suitability channel. We argue this reflects the already low inequality levels and mature technology systems of high-institution economies, which leave little room for further suitability-driven compression.

Fourth, however, when both institutional quality and human capital are simultaneously below median—a combination characteristic of low-income economies—the equalizing effect of suitability is significantly attenuated (coefficient on the double-low dummy interaction: +0.010, significant at 10 percent). This threshold pattern explains a well-documented anomaly: in low-income economies, capital-biased technical change often *widens* income dispersion even when nominally appropriate for the factor endowment (Jaumotte et al., 2013).

Mechanism analysis clarifies *how* these boundary conditions operate. Human capital amplifies suitability's equalizing effect primarily through global value chain (GVC) upgrading: suitability raises value-added production length more strongly where human capital is high, consistent with labor entering higher-skill GVC tasks. At the same time, high-human-capital economies show *less* expansion of vulnerable (informal, low-quality) employment in response to suitability improvements, suggesting that the equalizing mechanism shifts from quantity to quality of employment. Institutional quality, by contrast, operates through a consumer welfare channel: the interaction between suitability and institutions is positive and highly significant when the dependent variable is consumption-weighted total factor productivity, indicating that stronger institutions help convert efficiency gains into broadly shared purchasing power improvements. Together, these results reveal that human capital and institutional quality are *qualitatively different* boundary conditions: human capital is an absorption and amplification condition, while institutional quality is primarily a diffusion and floor condition.

Contributions. This paper makes three contributions to the literature. First, we extend the biased technical change literature—which has focused on the aggregate direction of innovation (Acemoglu, 2002; Acemoglu and Restrepo, 2018, 2020, 2022)—by introducing a boundary-condition framework that explains cross-country heterogeneity in the distributional consequences of factor-aligned technical change. Second, we complement studies of technology and inequality (Jaumotte et al., 2013; Krusell et al., 2000; Moll et al., 2022; Karabarbounis and

Neiman, 2014) by showing that the same variable—suitability—reduces inequality in human-capital-rich and institutionally strong economies but fails to do so when both conditions are absent simultaneously. Third, we provide cross-country evidence that GVC integration is a key mechanism linking human capital to the distributional benefits of technical change, adding to a growing literature on trade, technology, and inequality (Hornbeck and Moretti, 2024; Rodríguez and Rodrik, 2000; Timmer et al., 2015).

The remainder of the paper proceeds as follows. Section 2 develops the theoretical framework and states four testable hypotheses. Section 3 describes the data, variables, and empirical strategy. Section 4 presents the main regression results. Section 5 reports the mechanism analysis. Section 6 discusses robustness and limitations. Section 7 concludes with policy implications.

2 Theoretical Framework and Hypotheses

2.1 Biased Technical Change Suitability and Inequality

Acemoglu and Zilibotti (2001) show that technologies developed for high capital-labor-ratio economies raise productivity less when transplanted to labor-abundant developing countries. The key insight is that innovation embeds a factor bias: capital-intensive R&D produces blueprints that complement capital in production, and using such blueprints in capital-scarce economies wastes their comparative advantage. Li et al. (2026) and Li et al. (2024) formalize *suitability* as the realized alignment between factor bias and factor endowments: it equals the log marginal product ratio $\ln(\partial Y/\partial K) - \ln(\partial Y/\partial L)$, which captures how closely an economy's technical trajectory matches its actual capital-labor ratio.

When suitability is high, capital-biased technical change amplifies total factor productivity through three channels. *Channel 1 (GVC upgrading):* higher suitability enables firms to enter more capital-intensive, higher-value-added segments of global production networks, redistributing rents from foreign lead firms toward domestic labor. *Channel 2 (employment* *quality):* factor alignment shifts production toward tasks where labor and capital are complements rather than substitutes (Autor et al., 2003), raising the relative demand for middle-skill workers and attenuating wage polarization. *Channel 3 (welfare diffusion):* efficiency gains lower consumption costs and improve access to public goods, raising real incomes at the lower end of the distribution (Atkinson, 2015).

Hypothesis 1. *Capital-biased technical change suitability reduces income inequality.*

2.2 Human Capital as the Absorption Boundary

The three channels above are all contingent on workers being able to enter new tasks alongside capital. When human capital is high, workers possess the cognitive and technical skills to operate, adapt, and improve new equipment. Firms adopting capital-intensive technologies then need educated workers to manage and augment machines, raising skilled-labor demand and compressing the skill premium via *capital-skill complementarity* (Krusell et al., 2000). The GVC channel is especially human-capital-intensive: moving into higher-value-added production stages requires organizational capability, process engineering, and managerial depth that low-education labor forces lack (Goldin and Katz, 2009).

When human capital is low, the reverse applies. Workers cannot enter technologically sophisticated tasks, so capital-biased technical change raises capital returns without commensurate gains for labor. Tasks that could in principle be upgraded remain off-limits to the bulk of the workforce, and the employment margin shifts toward informal, low-quality jobs. The GVC channel stalls because firms cannot climb the value ladder without educated workers. Consequently, suitability's equalizing power depends critically on the economy's human capital stock.

Hypothesis 2. *Higher human capital amplifies the income-equalizing effect of capital-biased technical change suitability.*

2.3 Institutional Quality as the Diffusion Boundary

Even when suitability raises economy-wide productivity and human capital enables workers to enter new tasks, distributional benefits may not materialize if institutions are weak. Strong institutions—secure property rights, competitive markets, transparent governance, and robust social insurance—limit rent-seeking, protect workers' bargaining power, and channel public revenues from growth into health, education, and welfare (Acemoglu et al., 2005; Hall and Jones, 1999). In such environments, the efficiency gains from factor-aligned technical change flow broadly: firms compete away rents, governments can tax and redistribute, and workers have legal recourse against monopsonistic wage suppression.

Conversely, when institutions are weak, elites and politically connected firms can appropriate the surplus generated by suitability improvements (North, 1990). Tax systems are too thin to redistribute; public goods provision stagnates; labor regulations go unenforced. The consumer welfare channel is especially sensitive: turning aggregate efficiency gains into lower consumer prices and better public services requires fiscal capacity and rule of law.

Crucially, the moderation effect of institutional quality is unlikely to be linear. High-institution economies already exhibit compressed inequality and mature technological systems, so the marginal distributional gain from further suitability improvements is small. The critical role of institutions may be as a *floor condition*: preventing the worst distributional outcomes when suitability rises in economies where human capital is also insufficient.

Hypothesis 3. *Institutional quality acts as a threshold rather than a linear amplifier: the equalizing effect of suitability is significantly attenuated only when both institutional quality and human capital are simultaneously below the median.*

2.4 Channels of Boundary Conditions

The two boundary conditions engage distinct mechanisms. Human capital expands the set of workers who can perform the high-skill tasks associated with GVC upgrading, so the primary human-capital-mediated channel should be value-added GVC integration. Institutional quality, by contrast, determines whether efficiency gains are converted into public goods and lower consumer prices, so the primary institution-mediated channel should be consumer welfare improvement.

Hypothesis 4. *Human capital amplifies suitability's equalizing effect primarily through global value chain upgrading, while institutional quality operates primarily through consumer welfare diffusion.*

3 Data, Variables, and Empirical Strategy

3.1 Sample and Data Sources

Our sample consists of 64 economies observed annually from 1980 to 2019, yielding a balanced panel of 2,560 economy-year observations. The dataset is constructed by Li et al. (2026) and integrates five primary sources: (i) the Standardized World Income Inequality Database (SWIID) (Solt, 2016) for income inequality; (ii) Penn World Table version 10.0 (Feenstra et al., 2015) for output and factor inputs; (iii) the Fraser Institute's Economic Freedom Index (Gwartney et al., 2020) for institutional quality; (iv) historical schooling data from Clio Infra for the 1970 human capital proxy; and (v) the UIBE Global Value Chain Database for GVC indicators.

We extend the original empirical program of Li et al. (2026) by adding moderation specifications without altering the outcome variable, core explanatory variable, controls, or fixed-effect structure. This design maximizes comparability with the baseline estimates.

3.2 Variables

Income inequality ($\ln \text{Gini}_{it}$). The outcome variable is the natural logarithm of the market Gini coefficient—the pre-tax, pre-transfer Gini—from SWIID. Using market income keeps the measure responsive to labor and capital market outcomes before redistribution compresses it, making it more suitable for detecting the market-level distributional effects of technological change.

Technical change suitability ($\ln \mathrm{Suit}_{it}$). The core explanatory variable is the log ratio of the marginal product of capital to the marginal product of labor, constructed by Li et al. (2024). Formally:

$$\ln \mathrm{Suit}_{it} = \ln\Big(\mathrm{MPK}_{it}\Big) - \ln\Big(\mathrm{MPL}_{it}\Big), \tag{1}$$

where marginal products are derived from a CES production function estimated at the economy level. Higher values indicate that capital-biased technical change is better aligned with the economy's capital-labor endowment ratio.

Institutional quality (EFI_{it}). We use the Economic Freedom Index (EFI) from the Fraser Institute, which aggregates scores on property rights protection, monetary stability, trade openness, government size, and regulatory environment (scale 0–10). Higher EFI values indicate stronger market-supporting institutions.

Human capital ($\ln \mathrm{HC}_i$). We proxy human capital using the log of mean years of schooling in 1970, from Clio Infra (Li et al., 2026). This pre-sample measure is largely predetermined with respect to post-1980 inequality dynamics, reducing reverse-causation concerns. The cost is that it cannot capture within-sample human capital changes; we discuss this limitation in Section 6.

Control variables. We follow Li et al. (2026) in controlling for trade openness (tra, exports plus imports as a share of GDP), institutional quality (as above), the old-age dependency ratio (old), and log population ($\ln \mathrm{pop}$). All controls enter contemporaneously alongside economy and year fixed effects.

3.3 Descriptive Statistics

Table 1 reports descriptive statistics for all variables. The log Gini has a mean of 3.69 and a standard deviation of 0.28, reflecting substantial cross-country and over-time variation. Suitability averages 0.01 log points—close to the neutral benchmark—with a standard deviation of 0.48, spanning from -2.33 for highly labor-intensive economies to $+1.88$ for capital-intensive ones. Institutional quality averages 6.68 on a 0–10 scale, with a standard deviation of 1.26. The 1970 schooling proxy has a mean of 1.78 log years and standard deviation of 0.56, capturing large cross-country variation in historical human capital levels.

3.4 Empirical Strategy

Baseline specification. We estimate:

$$\ln \text{Gini}_{it} = \alpha + \beta \ln \text{Suit}_{it} + \boldsymbol{\gamma}' \mathbf{X}_{it} + \lambda_t + \theta_i + \varepsilon_{it}, \tag{2}$$

where λ_t are year fixed effects, θ_i are economy fixed effects, and $\mathbf{X}_{it}$ is the vector of controls. Two-way fixed effects control for time-invariant cross-economy heterogeneity (e.g. geography, colonial history) and common temporal shocks (global recessions, financial crises).

Moderation specifications. To test Hypotheses 2–4, we add an interaction term:

$$\ln \text{Gini}_{it} = \alpha + \beta_1 \ln \text{Suit}_{it} + \beta_2 \Big(\ln \text{Suit}_{it} \times \text{Boundary}_{it} \Big) + \boldsymbol{\gamma}' \mathbf{X}_{it} + \lambda_t + \theta_i + \varepsilon_{it}, \tag{3}$$

where Boundary is alternately: (a) the mean-centered EFI (institutional quality); (b) the mean-centered 1970 log schooling (human capital); (c) a dummy equal to one for economies above median country-mean human capital ("high-HC group"); and (d) a dummy equal to one for economies simultaneously below median in both institutional quality and human capital ("double-low"). A negative $\hat{\beta}_2$ implies that the boundary variable amplifies the equalizing effect; a positive $\hat{\beta}_2$ implies attenuation.

Inference. Standard errors follow Driscoll and Kraay (1998), which are consistent under arbitrary cross-sectional dependence and $MA(q)$ serial correlation; we set the Bartlett kernel bandwidth to three periods. This procedure is more conservative than cluster-robust standard errors and is better suited to panels with moderate T relative to N (Driscoll and Kraay, 1998). All continuous moderators are demeaned before constructing interaction terms so that the coefficient on $\ln \text{Suit}_{it}$ retains its interpretation as the average partial effect at the sample mean of the moderator.

4 Main Results

4.1 Baseline Effect and Boundary Conditions

Table 2 reports results from equations (2) and (3). Column (1) estimates the baseline specification without interactions. The coefficient on $\ln \text{Suit}$ is -0.0171 and significant at the 5 percent level (standard error 0.0078). Given that the standard deviation of $\ln \text{Suit}$ is 0.48, this implies that moving from the median to one standard deviation above the median in suitability is associated with a decline in the log Gini of $0.48 \times 0.0171 \approx 0.82$ percent— a meaningful reduction in market income dispersion. Hypothesis 1 is supported.

Column (2) adds the interaction between suitability and mean-centered institutional quality (EFI). The main effect of suitability remains negative and significant (-0.0202, $p < 0.05$), but the interaction coefficient is -0.0036 with a standard error of 0.0051, which is statistically indistinguishable from zero. The data do not support a linear amplification of suitability's equalizing effect by institutional quality. As discussed in Section 2, this null result is consistent with diminishing returns in high-institution economies: EFI independently reduces inequality ($\hat{\gamma}_{EFI} < 0$ in all specifications), but the marginal contribution of institutions to the suitability channel is not detectable at conventional significance levels.

Column (3) introduces the interaction with mean-centered 1970 human capital. The interaction coefficient is -0.0381, significant at the 1 percent level (standard error 0.0082).

This implies that when an economy moves from the sample mean to one standard deviation above in human capital (an increase of 0.56 log years), the equalizing effect of a one-unit increase in suitability rises by $0.56 \times 0.038 \approx 2.1$ percentage points. The suitability main effect remains strongly negative (-0.0351, $p < 0.01$), indicating robust equalizing effects even at mean human capital. Hypothesis 2 is supported.

Column (4) replaces the continuous human capital moderator with a high-HC dummy (equal to one for economies above median country-mean schooling in 1970). The interaction coefficient is -0.0244 ($p < 0.01$), confirming that the amplification is not an artifact of continuous scaling. High-human-capital economies experience an additional 2.4 percentage points of suitability-driven inequality reduction relative to low-human-capital economies.

Column (5) tests the double-low threshold in Hypothesis 3, replacing the moderator with a dummy for economies simultaneously below median in both institutional quality and human capital. The interaction coefficient is $+0.0100$, significant at the 10 percent level. In economies lacking both human capital and institutional quality, the equalizing effect of suitability is reduced by approximately 1 percentage point per unit of suitability—roughly one-third of the baseline effect. This supports the threshold interpretation of Hypothesis 3: the critical institutional role is not linear amplification but prevention of the worst distributional outcomes in doubly disadvantaged economies.

4.2 Economic Magnitude and Interpretation

The results in Table 2 reveal an asymmetry between the two boundary conditions. Human capital is both an amplifier (column 3) and a statistically sharper moderator than institutional quality (column 2). The double-low result (column 5) captures a threshold effect that the linear specification in column (2) misses. Together, these findings paint a coherent picture: *human capital is the absorption condition*— without it, the productivity gains from factor alignment cannot propagate into labor markets broadly—while *institutional quality is the diffusion floor*— without it, gains concentrate rather than spread, but the constraint is binding only when human capital is also weak.

Figure 1 illustrates the conceptual framework: suitability raises economy-wide productivity, but the translation into distributional improvement requires passing through human-capital and institutional filters. Both filters must be at least minimally operative to generate inclusive outcomes.

5 Mechanism Analysis

To test Hypothesis 4, we examine whether boundary conditions amplify suitability's effect through the three channels identified in Section 2: GVC upgrading, employment quality, and consumer welfare. We estimate equation (3) with alternative outcome variables, reporting results in Table 3.

5.1 Human Capital and GVC Upgrading

Columns (1) and (2) use GVC position (the ratio of forward to total GVC participation, a standard measure of upstream integration) and value-added production length (PLV, the average number of production stages in which a country participates) as outcomes.

Suitability significantly raises both GVC outcomes: the coefficient on GVC position is 0.084 ($p < 0.01$) and on PLV is 0.036 ($p < 0.10$), consistent with factor-aligned technical change enabling economies to enter more productive production stages. The interaction with human capital is positive and significant for PLV (0.054, $p < 0.01$) but insignificant for GVC position (0.015, $p > 0.10$). This suggests that human capital does not simply improve an economy's position in existing GVC stages but enables it to extend the breadth of value-added activities— a richer form of GVC integration that more directly benefits a wider set of workers.

5.2 Human Capital and Employment Quality

Columns (3) and (4) examine labor participation rate and vulnerable employment rate (the share of total employment in self-employment and contributing family work) as outcomes.

Suitability significantly raises both the participation rate (0.429, $p < 0.01$) and the vulnerable employment rate (1.029, $p < 0.01$). The positive coefficient on vulnerable employment may seem counterintuitive, but it reflects a transition effect: factor-aligned capital deepening initially draws workers from subsistence agriculture into market employment, which raises measured vulnerable employment before the formal-sector wage premium becomes large enough to restructure employment patterns.

The human capital interaction with both labor participation (-1.557, $p < 0.05$) and vulnerable employment (-4.048, $p < 0.01$) is strongly negative. This pattern is theoretically coherent: in high-human-capital economies, the employment margin of suitability improvement lies in *quality* (formal, higher-skill, higher-wage employment) rather than *quantity* (expanding any form of market participation). High-human-capital economies already have near-full labor force participation, so suitability improvements redirect workers from vulnerable toward formal employment rather than expanding the extensive margin. This compositional shift—fewer vulnerable jobs per unit of suitability—is precisely the distributional improvement that reduces inequality in high-human-capital economies.

5.3 Institutional Quality and Consumer Welfare

Column (5) uses consumption-weighted total factor productivity (CWTFP) as a proxy for consumer welfare, following the approach in Feenstra et al. (2015). Suitability significantly raises CWTFP (0.044, $p < 0.01$). The interaction with institutional quality is positive and significant (0.027, $p < 0.01$). This implies that stronger institutions substantially amplify the welfare gains from suitability improvements: a one-unit increase in EFI (roughly moving from the 25th to the 50th percentile of institutional quality) raises the suitability coefficient by 0.027, approximately 60 percent of the baseline suitability effect.

This institutional amplification of the welfare channel explains why institutional quality does not appear as a significant linear moderator in the inequality regressions of Table 2. Institutional quality primarily converts efficiency gains into consumer price reductions and public goods improvements—outcomes that show up in CWTFP but not directly in market Gini coefficients. The market Gini measures pre-tax, pre-transfer wage and capital income dispersion; welfare gains from cheaper consumption or better public services are largely invisible to that measure. This measurement mismatch rationalizes why institutions act as a threshold rather than a linear moderator of market-income inequality.

5.4 Synthesis

The mechanism results support Hypothesis 4 and reveal a clear division of labor between the two boundary conditions. Human capital is the amplification mechanism that converts technical suitability into higher-quality employment and deeper GVC integration, directly compressing wage dispersion. Institutional quality is the diffusion mechanism that converts aggregate productivity gains into consumer welfare, operating through a channel that market Gini statistics do not fully capture. Both mechanisms are consequential for inclusive growth, but they operate through distinct pathways and appear in different dependent variables. Policies targeting inequality through technology need to engage *both* channels simultaneously.

6 Robustness and Limitations

Alternative standard errors. The Driscoll-Kraay estimator is our preferred inference procedure, but we also verify results using two-way cluster-robust standard errors (clustering by economy and year). The main suitability coefficient and the human capital interaction remain significant at conventional levels; the institutional quality linear interaction remains insignificant; the double-low threshold interaction retains the correct sign with marginal significance.

Winsorized outcome and explanatory variables. To check that results are not driven by outliers, we re-estimate Table 2 with the log Gini and log suitability winsorized at the 1st and 99th percentiles. The human capital interaction coefficient changes from -0.038 to -0.035, remaining significant at 1 percent. The double-low interaction moves from $+0.010$ to $+0.009$, retaining 10 percent significance.

Lagged suitability. To reduce concerns about contemporaneous measurement error, we replace $\ln \text{Suit}_{it}$ with one-, two-, and three-year lags. All three lags produce negative, significant coefficients on the main suitability effect, and the human capital interaction remains negative and significant throughout. This lag structure also partially addresses reverse causation: the Gini in year t cannot affect suitability in year $t - 3$.

Instrumental variables. Li et al. (2026) instrument suitability using four-period-lagged log capital stock per worker, the 1970 log schooling proxy, a Bartik shift-share instrument, and the log mean suitability of other economies in the sample. We verify that the main suitability effect is robust to 2SLS estimation using these instruments. Constructing interaction instruments for the moderation analysis requires interacting the base instruments with the moderators; we confirm that human capital continues to significantly amplify the suitability–inequality relationship in the 2SLS framework for the IV specifications where sufficient first-stage strength is available.

Alternative human capital measures. The 1970 schooling proxy captures the historical stock but not within-sample human capital dynamics. When we use the PWT human capital index (available from 1990) as a robustness check on the post-1990 subsample, the human capital interaction remains negative and significant (-0.029, $p < 0.05$), and the mechanism results are qualitatively unchanged. This suggests our results are not driven by the particular construction of the human capital proxy.

Alternative institutional quality measures. The Economic Freedom Index emphasizes market-supporting institutions and may not capture redistribution capacity, labor protection, or public service provision. Replacing EFI with the World Bank Control of Corruption Index on the subsample where it is available does not generate a significant linear interaction, supporting the view that institutional quality acts as a threshold rather than a linear amplifier regardless of the specific institutional measure used.

Limitations. Several limitations merit acknowledgment. *First*, the balanced panel of 64 economies over-represents higher-income countries where panel data are available, which may bias estimates toward economies with relatively well-developed institutions and human capital. Extending the analysis to a broader, unbalanced panel including lower-income economies is a priority for future work. *Second*, cross-country identification conflates many country-level differences despite fixed effects. Within-country evidence using regional or industry-level data would provide sharper identification. *Third*, the pre-sample human capital proxy captures the historical basis for technical absorption but misses subsequent educational investments that may themselves respond to income inequality. Better instruments for the time-varying human capital moderator would strengthen causal interpretation. *Fourth*, the mechanism analysis is descriptive; the positive coefficient on vulnerable employment (before the human capital interaction) could reflect compositional changes rather than true labor market upgrading. Future work with household survey data can directly trace whether workers displaced from subsistence activities enter formal employment in economies with high human capital.

7 Conclusion

We examine whether and under what conditions capital-biased technical change suitability—the alignment between the direction of technical change and factor endowments—reduces income inequality across 64 economies from 1980 to 2019. Using two-way fixed effects estimation with cross-sectionally robust standard errors, we find that suitability significantly reduces inequality in the baseline specification. But this equalizing effect is conditional on two boundary factors.

Human capital is a quantitatively important amplifier: economies with higher 1970 schooling translate suitability improvements into deeper global value chain integration and higher-quality employment, substantially strengthening the compression of the wage distribution. Institutional quality does not linearly amplify the suitability–inequality channel, but it acts as an indispensable floor condition: when both institutions and human capital are simultaneously weak, the equalizing effect of suitability is significantly attenuated. Mechanism analysis reveals that human capital operates through GVC upgrading and employment quality, while institutional quality operates through consumer welfare channels not fully captured by market Gini statistics.

These findings carry concrete policy implications. *First*, technology-factor alignment—ensuring that innovation trajectories match economy-wide factor endowments—is a necessary but not sufficient condition for inclusive growth. Economies that import or develop capital-intensive technologies without simultaneously investing in labor force education will see productivity gains narrowly appropriated. *Second*, institutional reform and human capital investment are complementary but distinct priorities. Institutional quality matters most as a safeguard against extreme distributional failure; human capital is the active ingredient that converts technical suitability into broad-based wage gains. *Third*, global value chain strategy should be integrated with education policy. The finding that human capital amplifies the PLV channel suggests that upgrading industrial capabilities without a skilled workforce is unlikely to generate the distributional gains often attributed to GVC integration.

For rapidly industrializing economies—including China, where the empirical baseline originated—these results support policies that combine capital-intensive technological upgrading with sustained investment in education, vocational training, and social protection. The complementarity between "investing in things" and "investing in people" is not merely an equity objective; it is a prerequisite for the distributional dividends of technical progress to reach the broad population.

References

Acemoglu, Daron, "Directed Technical Change," *Review of Economic Studies*, 2002, *69* (4), 781–809.

_ **and Fabrizio Zilibotti**, "Productivity Differences," *Quarterly Journal of Economics*, 2001, *116* (2), 563–606.

_ **and Pascual Restrepo**, "The Race between Man and Machine: Implications of Technology for Growth, Factor Shares, and Employment," *American Economic Review*, 2018, *108* (6), 1488–1542.

_ **and** _ , "Robots and Jobs: Evidence from US Labor Markets," *Journal of Political Economy*, 2020, *128* (6), 2188–2244.

_ **and** _ , "Tasks, Automation, and the Rise in US Wage Inequality," *Econometrica*, 2022, *90* (5), 1973–2016.

_ **, Simon Johnson, and James A. Robinson**, "Institutions as a Fundamental Cause of Long-Run Growth," *Handbook of Economic Growth*, 2005, *1*, 385–472.

Atkinson, Anthony B., *Inequality: What Can Be Done?*, Cambridge, MA: Harvard University Press, 2015.

Autor, David H., Frank Levy, and Richard J. Murnane, "The Skill Content of Recent Technological Change: An Empirical Exploration," *Quarterly Journal of Economics*, 2003, *118* (4), 1279–1333.

Barro, Robert J. and Jong-Wha Lee, "International Measures of Schooling Years and Schooling Quality," *American Economic Review*, 1996, *86* (2), 218–223.

Caselli, Francesco and Wilbur John Coleman, "The World Technology Frontier," *American Economic Review*, 2006, *96* (3), 499–522.

Driscoll, John C. and Aart C. Kraay, "Consistent Covariance Matrix Estimation with Spatially Dependent Panel Data," *Review of Economics and Statistics*, 1998, *80* (4), 549–560.

Feenstra, Robert C., Robert Inklaar, and Marcel P. Timmer, "The Next Generation of the Penn World Table," *American Economic Review*, 2015, *105* (10), 3150–3182.

Goldin, Claudia and Lawrence F. Katz, *The Race between Education and Technology*, Cambridge, MA: Harvard University Press, 2009.

Gwartney, James, Robert Lawson, and Joshua Hall, "Economic Freedom of the World: 2020 Annual Report," Technical Report, Fraser Institute, Vancouver, BC 2020.

Hall, Robert E. and Charles I. Jones, "Why Do Some Countries Produce So Much More Output Per Worker than Others?," *Quarterly Journal of Economics*, 1999, *114* (1), 83–116.

Hornbeck, Richard and Enrico Moretti, "Estimating Who Benefits from Productivity Growth: Local and Distant Effects of City Productivity Growth on Wages, Rents, and Inequality," *Review of Economics and Statistics*, 2024, *106* (3), 587–607.

Jaumotte, Florence, Subir Lall, and Chris Papageorgiou, "Rising Income Inequality: Technology, or Trade and Financial Globalization?," *IMF Economic Review*, 2013, *61* (2), 271–309.

Karabarbounis, Loukas and Brent Neiman, "The Global Decline of the Labor Share," *Quarterly Journal of Economics*, 2014, *129* (1), 61–103.

Krusell, Per, Lee E. Ohanian, José-Víctor Ríos-Rull, and Giovanni L. Violante, "Capital-Skill Complementarity and Inequality: A Macroeconomic Analysis," *Econometrica*, 2000, *68* (5), 1029–1053.

Li, Xiaoke, Shanshan Zheng, Shuqing Zhang, and Huaiqi Shi, "Measurement and Structural Decomposition of Biased Technical Change Suitability," *Statistical Research*, 2024, (7). In Chinese.

Li, Xiaoping, Zhipeng Xu, and Xiaoke Li, "Toward Global Shared Prosperity: How Does the Suitability of Biased Technical Change Mitigate Income Inequality," *China Industrial Economics*, 2026, (4). In Chinese.

Moll, Benjamin, Lukasz Rachel, and Pascual Restrepo, "Uneven Growth: Automation's Impact on Income and Wealth Inequality," *Econometrica*, 2022, *90* (6), 2645–2683.

North, Douglass C., *Institutions, Institutional Change and Economic Performance*, Cambridge: Cambridge University Press, 1990.

Rodríguez, Francisco and Dani Rodrik, "Trade Policy and Economic Growth: A Skeptic's Guide to the Cross-National Evidence," *NBER Macroeconomics Annual*, 2000, *15*, 261–325.

Solt, Frederick, "The Standardized World Income Inequality Database," *Social Science Quarterly*, 2016, *97* (5), 1267–1281.

Timmer, Marcel P., Erik Dietzenbacher, Bart Los, Robert Stehrer, and Gaaitzen J. de Vries, "An Illustrated User Guide to the World Input-Output Database: The Case of Global Automotive Production," *Review of International Economics*, 2015, *23* (3), 575–605.

Tables

Table 1: Descriptive Statistics

Variable	Description	Obs.	Mean	Std. Dev.	Min	Max
$\ln \text{Gini}$	Log income inequality (market Gini)	2,560	3.6884	0.2847	2.8745	4.3449
$\ln \text{Suit}$	Log biased TC suitability	2,560	0.0103	0.4799	-2.3347	1.8755
EFI	Institutional quality (Fraser EFI)	2,472	6.6795	1.2648	3.1100	9.1100
$\ln \text{HC}_{70}$	Human capital (log schooling, 1970)	2,280	1.7800	0.5634	0.0951	2.5006
old	Old-age dependency ratio	2,560	0.1416	0.0813	0.0273	0.4852
tra	Trade openness (% GDP)	2,526	81.44	63.06	9.14	442.62
$\ln \text{pop}$	Log population (millions)	2,560	2.5210	1.7846	-1.4773	7.2681

Notes: Sample is a balanced panel of 64 economies, 1980–2019. TC = technical change. EFI = Economic Freedom Index (Fraser Institute, scale 0–10). Obs. for EFI and human capital differ because of missing values in the original sources.

Table 2: Capital-Biased Technical Change Suitability and Income Inequality: Boundary Conditions

	(1) Baseline	(2) Institution moderator	(3) Human capital moderator	(4) High-HC group	(5) Double-Low group
$\ln \text{Suit}$	-0.0171^{**}	-0.0202^{**}	-0.0351^{***}	-0.0139^{*}	-0.0270^{***}
	(0.0078)	(0.0082)	(0.0108)	(0.0083)	(0.0102)
$\ln \text{Suit} \times \text{EFI}$		-0.0036			
		(0.0051)			
$\ln \text{Suit} \times \ln \text{HC}_{70}$			-0.0381^{***}		
			(0.0082)		
$\ln \text{Suit} \times \mathbf{1}[\text{High-HC}]$				-0.0244^{***}	
				(0.0082)	
$\ln \text{Suit} \times \mathbf{1}[\text{Double-Low}]$					0.0100^{*}
					(0.0056)
Observations	2,438	2,438	2,158	2,158	2,158
Two-way FE	Yes	Yes	Yes	Yes	Yes
Controls	Yes	Yes	Yes	Yes	Yes

Notes: Dependent variable is $\ln \text{Gini}_{it}$ (log market Gini coefficient). All regressions include economy and year fixed effects. Controls are trade openness, institutional quality (EFI), old-age dependency ratio, and log population. EFI and $\ln \text{HC}_{70}$ are demeaned before constructing interaction terms. $\mathbf{1}[\text{High-HC}]$ equals one for economies above the median of country-mean 1970 schooling. $\mathbf{1}[\text{Double-Low}]$ equals one for economies simultaneously below median in both EFI and 1970 schooling. Standard errors in parentheses are Driscoll–Kraay (Driscoll and Kraay, 1998) with Bartlett kernel bandwidth of 3, robust to cross-sectional dependence and serial correlation. *, **, *** denote significance at the 10%, 5%, and 1% levels, respectively.

Table 3: Mechanism Analysis: Through Which Channels Do Boundary Conditions Operate?

	(1) GVC Position	(2) Value-Added Production Length	(3) Labor Particip. Rate	(4) Vulnerable Employment Rate	(5) Consumer Welfare (CWTFP)
$\ln \text{Suit}$	0.0839^{***}	0.0361^{*}	0.4291^{***}	1.0287^{***}	0.0436^{***}
	(0.0217)	(0.0217)	(0.1478)	(0.3246)	(0.0074)
$\ln \text{Suit} \times \ln \text{HC}_{70}$	0.0151	0.0539^{***}	-1.5574^{**}	-4.0478^{***}	
	(0.0281)	(0.0140)	(0.6397)	(0.7131)	
$\ln \text{Suit} \times \text{EFI}$					0.0268^{***}
					(0.0055)
Observations	888	888	1,513	1,585	2,438
Two-way FE	Yes	Yes	Yes	Yes	Yes
Controls	Yes	Yes	Yes	Yes	Yes

Notes: Dependent variables are as listed in column headers. GVC Position measures a country's forward-linkage share in global value chains (upstreamness index). Value-Added Production Length (PLV) measures the average number of production stages in which the economy participates. Labor Participation Rate and Vulnerable Employment Rate are expressed in percentage points. Consumer Welfare (CWTFP) is the consumption-weighted total factor productivity index from Penn World Table 10.0 (Feenstra et al., 2015). All regressions include economy and year fixed effects. Controls are trade openness, institutional quality (EFI), old-age dependency ratio, and log population. Continuous moderators are demeaned. Standard errors in parentheses are Driscoll–Kraay (Driscoll and Kraay, 1998) with bandwidth 3. *, **, *** denote significance at the 10%, 5%, and 1% levels, respectively.

[Figure: Conceptual diagram of suitability boundary conditions]

Suitability $\longrightarrow$ Productivity gains

$\xrightarrow{\text{Human Capital Filter}}$ GVC upgrading $\longrightarrow$ Lower inequality

$\xrightarrow{\text{Institution Filter}}$ Consumer welfare $\longrightarrow$ Lower *effective* inequality

When both filters are weak: gains concentrate $\Rightarrow$ Higher inequality

Figure 1: Conceptual Framework: Boundary Conditions on the Suitability–Inequality Channel